

Database Systems — Development Fixture

Database normalization reduces redundancy and improves data integrity.

A primary key uniquely identifies each row in a relation.

First normal form requires atomic attribute values and no repeating groups.

Second normal form removes partial dependencies on a composite candidate key.

Third normal form removes transitive dependencies of non-key attributes.

A transaction groups operations into a logical unit of work.

ACID means atomicity, consistency, isolation and durability.

Atomicity ensures either all transaction operations succeed or none do.

A database index can speed up retrieval at the cost of storage and writes.

Practice: explain how normalization prevents update anomalies.